

EN.601.315/415/615

Databases

Set 8 – Application Development

SKS – from Chapter 9

Some slides modified from Silberschatz, Korth, Sudarshan

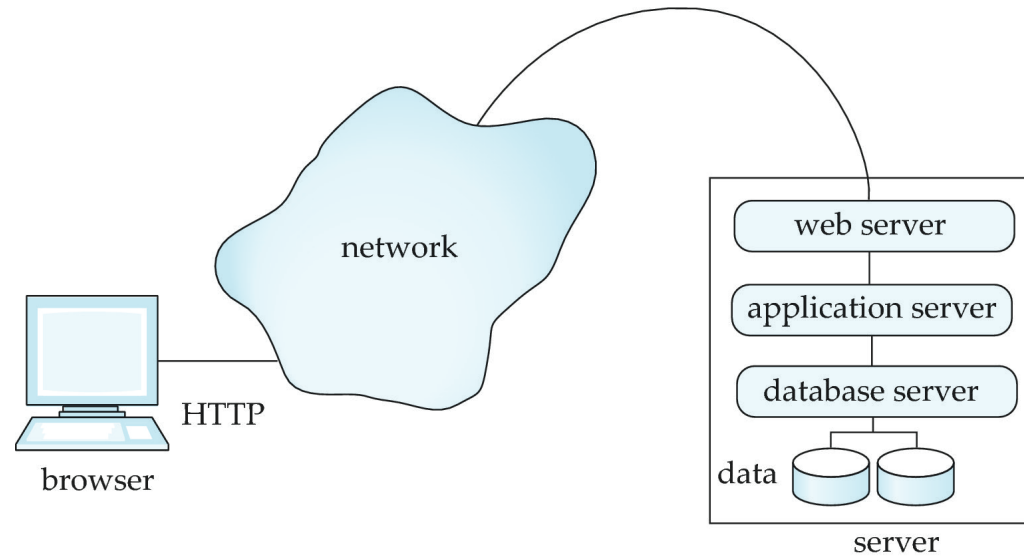
Goals

- Application Development with Database back-end
 - N-tier architecture
 - HTML
 - PHP
 - PHP + MySQL

Application Programs and User Interfaces

- Most database users do *not* use a query language like SQL
- Application programs
 - front-end
 - middle layer
 - backend

Three-Layer Application Architecture



Sample HTML

<https://www.w3schools.com/html/default.asp>

```
<!DOCTYPE html>
<html>
<head>
<title>Sample table and form</title>
    <style>
        table, th, td{border: 1px solid black}
    </style>
</head>
<body>
    <h1>Sample HTML Page</h1>
    <p>This page shows a few HTML elements:
        <ul>
            <li><a href = "https://www.google.com">Google</a></li>
            <li><a href = https://www.cs.jhu.edu>JHU CS Dept</a></li>
            <li><a href = "demo1.html">HTML demo</a></li>
        </ul>
    </p>
```

Sample HTML (cont)

```
<table>
  <tr> <th>ID</th> <th>Name</th> <th>Department</th> </tr>
  <tr> <td>00128</td> <td>Zhang</td> <td>Comp. Sci.</td> </tr>
  <tr> <td>12345</td> <td>Shankar</td> <td>Comp. Sci.</td> </tr>
</table>
<form action="PersonQuery" method="get">
  Search for: <select name="persontype">
    <option value="student" selected>Student </option>
    <option value="instructor"> Instructor </option>
  </select>
  <br>Name: <input type="text" size="20" name="name"><input type="submit" value="submit">
</form>
</body> </html>
```

Display of Sample HTML Source

Sample HTML Page

This page shows a few HTML elements:

- [Google](#)
- [JHU CS Dept](#)
- [HTML demo](#)

ID	Name	Department
00128	Zhang	Comp. Sci.
12345	Shankar	Comp. Sci.

Search for:

Name:

Server-Side Scripting

- Server-side scripting simplifies the task of connecting a database to the Web
 - HTML + code
 - Input values from HTML forms can be used directly in the embedded code/SQL queries.
 - Server executes code, generates the HTML
- Server-side scripting languages:

CGI Script Basics

- Common Gateway Interface (CGI)
 - “Common”: Not specific to any operating system or language
- Output file generated at runtime:
 1. When a program executed as a CGI script, “standard output” is redirected to web server
 2. Web server then redirects output to client's browser

How can CGI get data from user?

- Technique #1: Forms
- User enters data via a form, submits
- Browser requests the CGI file by name specified on form _____
- Web server **runs** the CGI program with form's data passed to it
- Script receives data in one of two ways:
 1. method = "get"
 2. method = "post"
- Use language-specific method to get these inside CGI program
- Technique #2: URL with parameters
- `
Seminars `

PHP Overview

<https://www.w3schools.com/php/default.asp>

- PHP tags
- Statements
- Variables
- Strings
- Accessing form variables

Sample PHP Script

```
<html>
  <head> <title> Hello </title> </head>
  <body>
    <?php
      if (!isset($_GET['name']))
      { echo "Hello World"; }
      else { echo "Hello, " + $_GET['name']; }
    ?>
  </body>
</html>
```

PHP Functions and Arrays

```
<?php require_once('arrayFunctions.inc.php');
```

```
$content = '<p>Working with Arrays</p>';
```

```
$array = array('thing1', 'thing2');  
$array[2] = 'thing3';
```

```
$content .= '<p>'.displayArray($array) . '</p>';
```

```
$prices = array('Tires' => 100, 'Oil' => 10);  
$prices['Spark plugs'] = 200;
```

```
$content .= '<p>The values in $prices array are <br />' . displayAssociativeArray($prices) . '</p>';  
sort($array); //sort array  
$content .= '<p> Sorted array '.displayArray($array) . '</p>';
```

```
ksort($prices); //sort associative array by key  
$content .= '<p>The values in $prices array sorted by key are <br />' . displayAssociativeArray($prices) . '</p>';  
asort($prices); //sort associative array by value  
$content .= '<p>The values in $prices array sorted by value are <br />' . displayAssociativeArray($prices) . '</p>';  
?>  
<!DOCTYPE html> ...
```

```
<?php echo $content; ?>  
</body></html>
```

array_functions.inc.php

```
<?php
```

```
//return a string to display an array
```

```
function displayArray($array){  
    $result = '<p>';  
    for($i = 0; $i < count($array); $i++){  
        $result .= "Element at index $i is $array[$i] <br />";  
    }  
    $result .= '</p>';  
  
    return $result;  
  
}
```

```
//return a string to display an associative array
```

```
function displayAssociativeArray($array){  
    $result = '<p>';  
    foreach($array as $key => $value){  
        $result .= "Element with key $key has value $value <br />";  
    }  
    $result .= '</p>';  
    return $result;  
  
}  
?>
```

PHP and MySQL

- <https://www.php.net/manual/en/book.mysql.php>
- mysql library

Establishing a Connection

```
$db = new mysqli( 'dbase.cs.jhu.edu',  
                'your_dbase_username',  
                'your_dbase_password',  
                'your_dbase_database_name' );  
  
if ( $db->connect_error ) {  
    die( "Connection failed: " . $db->connect_error ); }
```

- Better: store in separate file (mysql.inc.php) and then include when needed

Closing a connection

- `$db->close()`

Perform a Simple Query

```
$db->query("DROP TABLE IF EXISTS test");
$db->query("CREATE TABLE test(id INT)");
$db->query("INSERT INTO test(id) VALUES (1), (2), (3)");

$result = $db->query("SELECT id FROM test ORDER BY id ASC");

echo "<p>Result set:";
foreach ($result as $row) {
    echo " id = " . $row['id'] . "<br>\n";
}
echo "</p>";
```

Perform a Parameterized Query

- `$query = 'SELECT course_id, title, dept_name FROM section JOIN course USING (course_id) WHERE semester=? AND year = ?';`
- `$stmt = $db->prepare($query);`
- `$semester = $_GET['semester']; $year = $_GET['year'];`
- `$stmt ->bind_param('si', $semester,$year);`

Perform a Parametrized Query

- `$success = $stmt->execute();`
- ```
if (!$success) {
 echo "<h5>ERROR: " . $db->error .
 " for query *$query*</h5><hr>";
}
```

# Process Results – Fetch Rows

- `$result = $stmt->get_result();`
- How many rows (results) did the query return?

- `$result->num_rows;`

- Get 1 row at a time

```
while($row = $result->fetch_assoc()) {
 ...$row["col1"], $row["col2"]
}
```

- Get all the rows at once

```
$rows = $result->fetch_all(MYSQLI_ASSOC);
foreach($rows as $row)...
```

# Retrieve Response Rows – Bind Result

- We need to bind each column to a variable.
  - `$stmt->bind_result($course, $title, $dept_name);`

- Then retrieve the rows

```
while ($stmt->fetch()) {
 echo "$course $title $dept_name
";
}
```

# Useful Functions

- `$stmt->store_result();`
- `$db->affected_rows;`
- `$num_rows = $stmt->num_rows;`
- `$insert_id = $stmt->insert_id;`

# Exercise

- Build your **mysql.inc.php** file
- Build a PHP script `displayGrades.php` that will:
  - Connect to the db
  - Display as an HTML table:
    - **ID, name, course\_id**, and **grade** for all students and courses
      - `student(ID, name, dept_name, tot_credits)`
      - `takes(ID, course_id, sec_id, semester, year, grade)`
- Create a form with an input field for student ID. When form is submitted, `displayGrades.php` should be called
- Modify previous script so that, if an ID is provided from the form, it:
  - Takes the ID as input into the query
  - Displays the ID, name, course\_id, grade for that student



# HTML and PHP on CS.JHU machines

- All your source codes should be located in the ~/public\_html folder
- If there isn't create one as follows:
  - >**mkdir public\_html**
  - >**chmod 705 public\_html**
- You have to change the permission of a file if you want to access it via web-browser. (IMPORTANT STEP)
  - >**chmod 704 demo1.php**
- To test for syntax errors:
  - >**php -l demo1.php**
- In your php code:
  - error\_reporting(E\_ALL);
  - ini\_set('display\_errors', 1);
  - ini\_set('display\_startup\_errors', 1);
- URL to access: [https://www.cs.jhu.edu/~your\\_cs\\_username/your\\_filename](https://www.cs.jhu.edu/~your_cs_username/your_filename)

# Calling Stored Procedures

```
require_once('mysql.inc.php');
```

```
$db->query("DROP PROCEDURE IF EXISTS p");
```

```
$db->query('CREATE PROCEDURE p() READS SQL DATA BEGIN SELECT id FROM test; SELECT id + 1 FROM test; END;');
```

```
$db->real_query("CALL p()");
```

```
do {
 if ($result = $db->store_result()) {
 echo "results:
";
 while ($row = $result->fetch_array()){
 echo "value ". $row[0]. "
";
 }
 $result->free();
 }
} while ($db->next_result());
```

# Calling Stored Procedure with IN and OUT

```
Require_once('mysql.inc.php');

$db->query("DROP PROCEDURE IF EXISTS strLen");
$db->query('CREATE PROCEDURE strLen(IN myStr VARCHAR(20), OUT charCount INT) BEGIN SET charCount = LENGTH(myStr); END;');

$str = "";
if (isset($_REQUEST['str'])){
 $str = $_REQUEST['str'];
}

$db->query("CALL strLen('$str', @len)");
$result = $db->query("SELECT @len as len_out");
$row = $result->fetch_assoc();

echo "<p>Length of string *$str* is ". $row['len_out'] . "</p>";
```

# Exercise – Stored Procedure

- Create a stored procedure StudentGradesSP
  - Input parameter: ID
  - Output parameter: GPA
    - Lists **ID**, **name**, **course\_id**, and **grade** for all courses taken by the student
      - student(ID, name, dept\_name, tot\_credits)
      - takes(ID, course\_id, sec\_id, semester, year, grade)
      - gradePoint(letterGrade, pointValue)
- Modify displayGrades.php to
  - call StudentGradesSP
  - display GPA
  - in addition to table with ID, name, course\_id, and grade

# Sessions in PHP – tracking info across webpages

- `session_start()`
- `$_SESSION['myvar'] = 'some value'`
- `session_start()`
- `if ( isset($_SESSION['myvar']) ) { //OK code }`
- `unset($_SESSION['myvar'])`
- `$_SESSION= array()`
- `session_destroy()`

# Demo - sessions

# Exercise – fill in the blanks

```
require_once('page.inc.php');
require_once('myConnectDB.inc.php');
require_once('user.inc.php');
session_start();
//create a new page object
$page = new Page("Login page");
//if page already submitted, process info
if (isset($_POST['login'])){
 //get posted user id and password

 //connect to the db

 //call the validate function in user class
```

```

/
 /if everything OK, set session variables
 //and go to secret page
 if (
){

 header('Location: secretPage.php');

 }
 //else, display an error message and redisplay the input form
 else{

 }

}
//set the content: a form to get the login information
$page->content .=
'<form action = "loginPage.php" method = "post">
<p><label>Username: <input type="text" name="username" /></label></p>
<p><label>Password: <input type="password" name="password"/></label></p>
<p><input type="submit" name = "login" value = "Login"/>
</form>';
//display the page
$page->display();
?>

```